E4-20
E9AC3

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2013

MSc and EEE PART IV: MEng and ACGI

FULL CUSTOM INTEGRATED CIRCUIT DESIGN

Friday, 10 May 2013 10:00am

Time allowed: 2 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible — First Marker(s): T.G. Constandinou
Second Marker(s): C. Mehring

Information for Invigilators:

Students may bring any written or printed aids to the examination.

Information for Candidates:

1. Students may need a ruler and coloured pens (red, green, blue and black).

2. Use the following expressions where appropriate:

MOSFET drain current: $$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}\left(V_{GS} - V_{Th}\right)^2$$

Standard deviation of threshold voltage mismatch: $$\sigma\left(\Delta V_{Th}\right) = \frac{AV_{Th}}{\sqrt{WL}}$$

3. For the expressions above, use the following parameters for 0.18µm CMOS Technology:

$\mu_n C_{ox} = 200\mu A/V^2$
$\mu_p C_{ox} = 80\mu A/V^2$
$V_{Th(NMOS)} = 0.4V$
$V_{Th(PMOS)} = -0.5V.$
$AV_{Th(NMOS)} = 4.0mV\mu m$
$AV_{Th(PMOS)} = 4.5mV\mu m$

1. A full-custom layout design is shown below in Figure 1.1 for an integrated circuit implemented in a commercially available 0.18µm 1P6M CMOS technology (only 2 metal layers used in design).

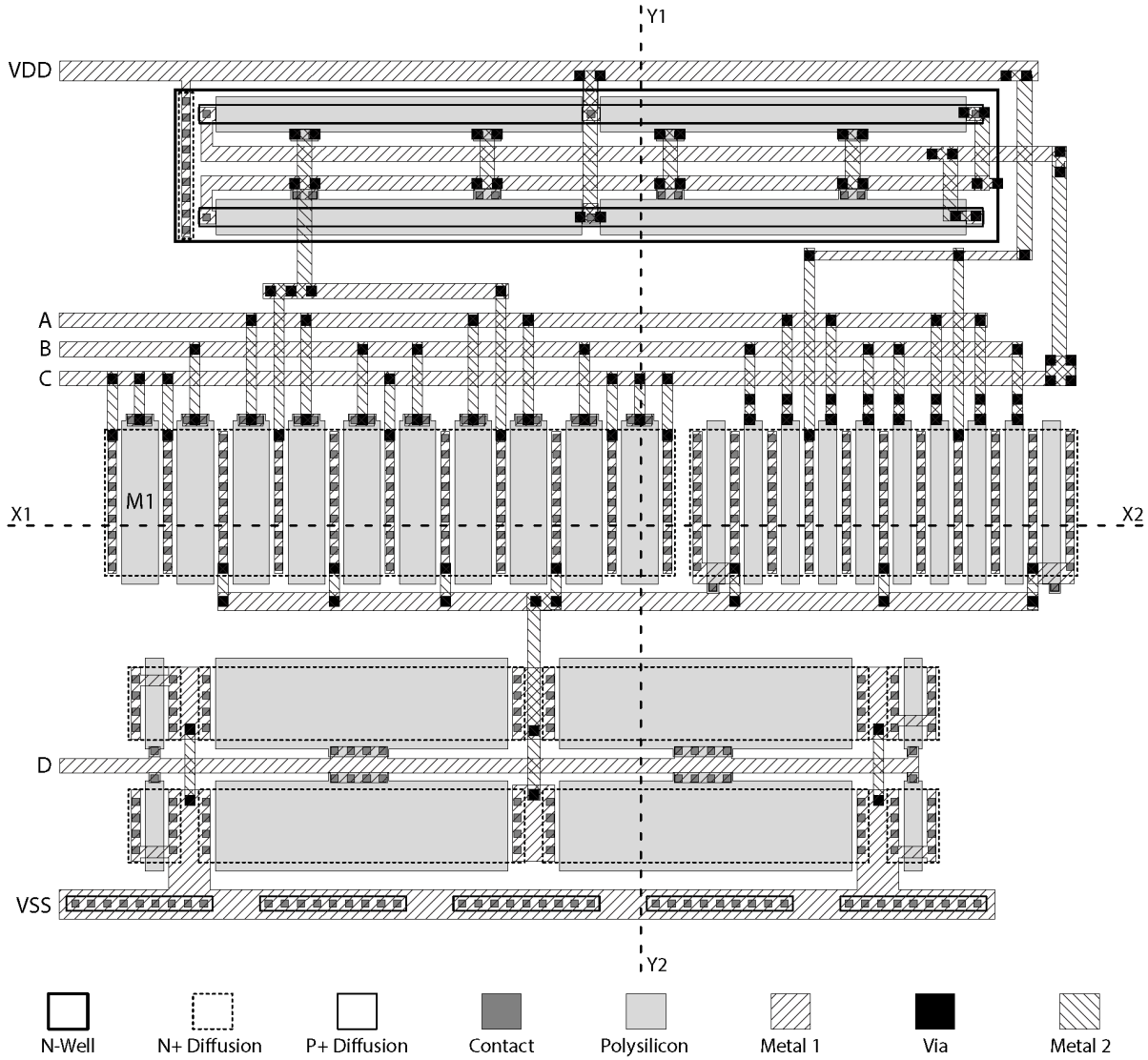


Figure 1.1

a) Extract and draw the transistor-level circuit schematic for this design showing the nodes VDD, VSS, A, B, C, D.

Given that the transistor labelled M1 is of size W/L = (4µm/1µm), label all transistors in this circuit with their sizes. Hint: Several of the devices are constructed use a multi-fingered layout (i.e. using multiple identical devices connected in parallel).

[10 marks]

b) Draw the vertical cross-sections along the lines X1-X2 and Y1-Y2. Label your diagrams indicating the n-well region and the different structures (eg. Metal1, Metal2), types and levels of doping (eg. P+, N+, etc).

[8 marks]

c) What function does this circuit perform?

[2 marks]

2. The cross-section of commercially available 0.35µm 2P3M CMOS technology is shown below in Figure 2.1.

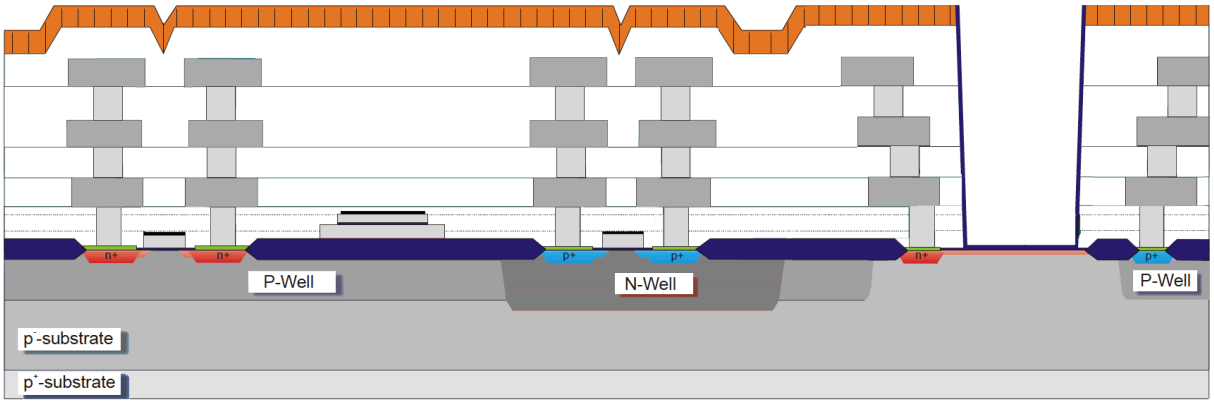


Figure 2.1

a) List 8 different devices (i.e. either passive or active integrated components) that can be fabricated in this technology. For each device, sketch its cross-section (i.e. structure) and comment on any limitations.

[12 marks]

b) Given that in this technology a 3T APS Pixel would occupy a 5µm x 5µm footprint, a 25mm^2 die size is required per megapixel (excluding column amplifiers, data converter and any other circuits outside the imaging array). How would this imager architecture scale to a 32nm CMOS technology? What (if any) are the limitations in scaling image sensors?

[4 marks]

c) Using the fact that the penetration depth of light in Silicon reduces with wavelength (by the Beer Lambert law), describe how a standard CMOS technology can be exploited to provide spectrally selective photodiodes (without colour filters).

[4 marks]

3. The design of a N-bit binary-weighted current-steering Digital-to-Analogue (DAC) converter shown below in Figure 3.1 is to be implemented in a standard 0.18µm CMOS technology.

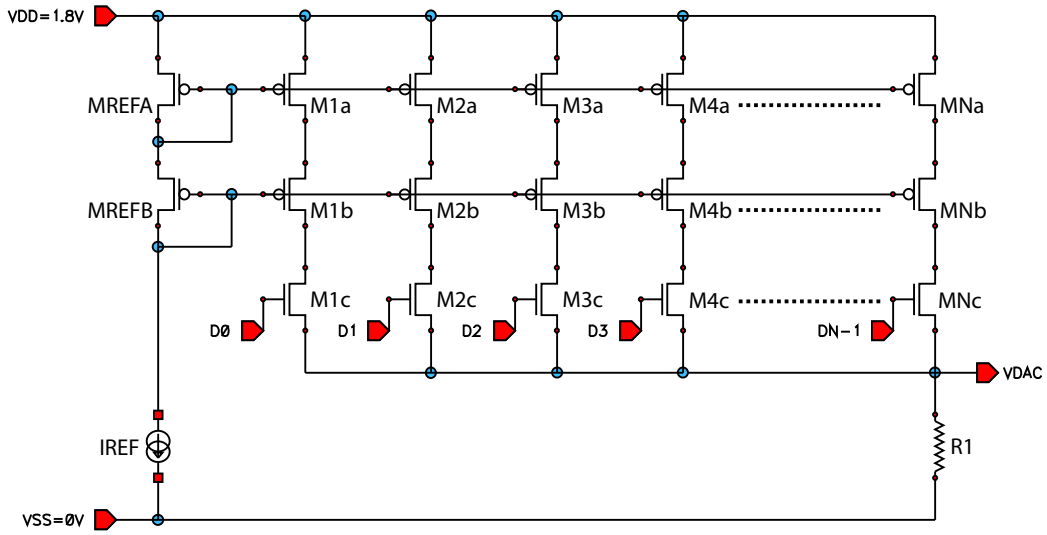


Figure 3.1

For this question, use the expressions provided in page 1 where appropriate.

a) Given that N=8, I_{REF}=2µA and R1=10KΩ calculate the ratio between device aspect ratios (i.e. W/L's) for MREFa:M1a:...M8a given that a 1V output range is required.

[4 marks]

b) Sketch how you would arrange the critically matched devices (i.e. current mirrors) in a layout design to improve the matching. Show only the devices corresponding to the 4 LSB's (i.e. D0, D1, D2 and D3).

Note that a detailed layout is not required – you are only required to sketch the arrangement (i.e. order/orientation and placement of the devices).

[4 marks]

c) Compare this DAC topology to that of an R-2R ladder. State (and explain) two advantages of each topology compared to the other.

[4 marks]

d) Given that $(W/L)M_{REFA}=(W/L)M_{REFB}$=25µm/25µm, and assuming that all devices are in strong inversion (and saturation), determine the worst case for the *Effective Number Of Bits (ENOB)* of the converter.

[8 marks]

4. The design of a clock generation circuit (i.e. oscillator) is shown below in Figure 4.1. This uses a Schmitt trigger to alternate between charging and discharging a capacitor using a constant current source/sink. The current generator is based on a beta-multiplier reference.

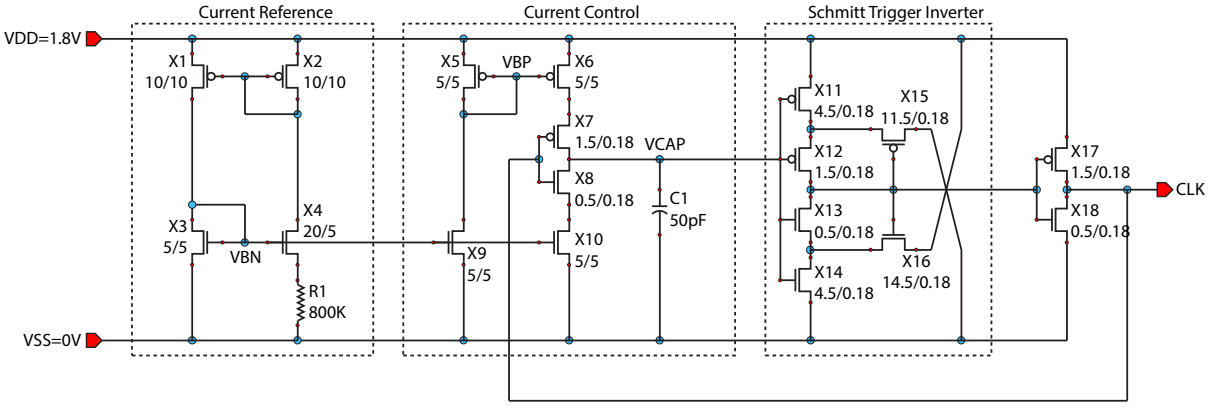


Figure 4.1

For this question, use the expressions provided in page 1 where appropriate.

a) Assuming devices X1-X4 are all in saturation and operating in strong inversion, derive an expression for I_{REF} in terms of component parameters, for example: R, $(W/L)_1$, etc. and show that this is independent of power supply.

[5 marks]

b) Using your answer to (a), evaluate the bias current (I_{REF}).

[2 marks]

c) Determine the switching points (V_{SLH} and V_{SHL}) of the Schmitt trigger inverter.

[5 marks]

d) Determine the oscillation frequency.

[2 marks]

e) Given that this will be implemented in a 0.18µm CMOS technology with MIM capacitors and a high resistance poly module, estimate the silicon area required to implement this circuit. Use the following technology parameters: C_{MIM}=0.7fF/µm^2, R_{POLYH}=1KΩ/□.

[4 marks]